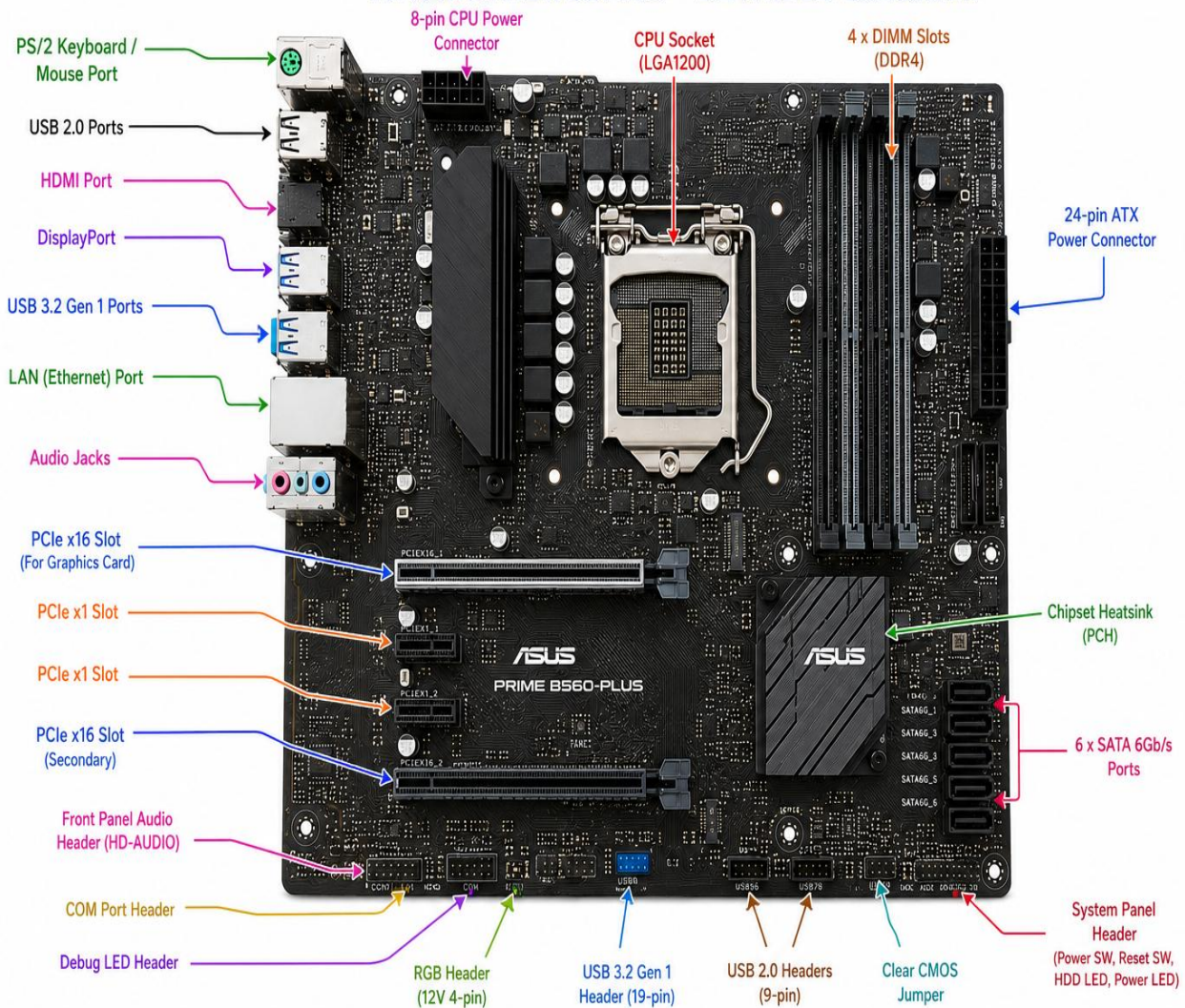


ASSIGNMENT -1.2

Q1. Draw a labelled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).

ATX MOTHERBOARD – DETAILED DIAGRAM



Q2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

❖ **CPU Socket Type: -**

➤ **LGA1151:**

- Intel desktop socket used for 6th, 7th, 8th, and 9th Gen Intel Core processors (depending on chipset).
- **CPU Model: Intel Core i7-9700K**

➤ **AM4:**

- AMD desktop socket supporting Ryzen processors from 1st to 5th Generation.
- **CPU Model: AMD Ryzen 5 5600X**

➤ **LGA1200:**

- Intel desktop socket used for 10th and 11th Gen Intel Core processors.
- **CPU Model: Intel Core i5-10400.**

Q3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

- The **CPU** is the brain of the computer. It does all the hard thinking, calculations, and runs the programs.
 - It does this by continuously repeating a 4-step cycle called the **Fetch-Decode-Execute Cycle**:
 - The **Chipset** is the traffic police on the motherboard. It links the CPU to everything else, like the storage drive, RAM, internet card, and speakers.
 - The **primary function** of a chipset is to act as the **communications hub** or **traffic controller** on the motherboard.
 - If the CPU is the brain, the chipset is the nervous system. The CPU cannot talk to most parts of your computer directly—it needs the chipset to route the data.
- Here are the 3 main things a chipset does:
1. **Connects the CPU to Everything Else**
 2. **Controls Data Flow**
 3. **Determines Motherboard Features**

How they work together when you open Spotify:

- **Click Spotify:** The CPU hears your click and tells the Chipset, "**Go grab Spotify from the storage drive.**"
- **Loading the App:** The Chipset moves the Spotify files from your slow storage drive into the fast memory (RAM).

- **Running the App:** The CPU reads the files from the RAM and displays the Spotify screen.
- **Playing Music:** When you click play, the Chipset grabs the song from the internet. The CPU processes the music data.
- **Making Sound:** The CPU sends the sound signals back through the Chipset to your speakers so you can hear the music.

Q4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labelling each one.

MOTHERBOARD:

